Homework Assignment 6

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Rutgers University

Jeffrey Lu

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Chapter 1.2, Question 2:

Suppose $A = \{\pi, e, 0\}$ and $B = \{0, 1\}$.

(a)

$$A \times B = \{(\pi, 0), (\pi, 1), (e, 0), (e, 1), (0, 0), (0, 1)\}.$$

(b)

$$B \times A = \{(0, \pi), (0, e), (0, 0), (1, \pi), (1, e), (1, 0)\}.$$

(c)

$$A \times A = \{(\pi, \pi), (\pi, e), (\pi, 0), (e, \pi), (e, e), (e, 0), (0, \pi), (0, e), (0, 0)\}.$$

Chapter 1.2, Question 8:

$$\begin{aligned}
\{0,1\}^4 &= \{0,1\} \times \{0,1\} \times \{0,1\} \times \{0,1\} \\
&= \{(0,0), (0,1), (1,0), (1,1)\} \times \{0,1\} \times \{0,1\} \\
&= \{(0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,0), (1,0,1), (1,1,0), (1,1,1)\} \times \{0,1\} \\
&= \{(0,0,0,0), (0,0,0,1), (0,0,1,0), (0,0,1,1), (0,1,0,0), (0,1,0,1), \\
&\quad (0,1,1,0), (0,1,1,1), (1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1), \\
&\quad (1,1,0,0), (1,1,0,1), (1,1,1,0), (1,1,1,1)\}.
\end{aligned}$$

Chapter 1.2, Question 18:

Chapter 1.4, Question 2:

$$\begin{aligned}\mathcal{P}(\{1,2,3,4\}) = \{&\emptyset, \{1\}, \{2\}, \{3\}, \{4\},\\ &\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\},\\ &\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\},\\ &\{1,2,3,4\}\}.\end{aligned}$$

Chapter 1.4, Question 12:

$$\{X \in \mathcal{P}(\{1,2,3\}) : 2 \in X\} = \{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}.$$

Chapter 1.4, Question 18:

Suppose $|A| = m$ and $|B| = n$. Then,

$$\begin{aligned}|\mathcal{P}(A \times \mathcal{P}(B))| &= 2^{|A| \cdot 2^{|B|}}\\ &= 2^{m2^n}\end{aligned}$$

Chapter 1.8, Question 10:

(a)

$$\bigcup_{x \in [0,1]} [x,1] \times [0,x^2] = \{(a,b) \in [0,1]^2 : b \leq a^2\}.$$

(b)

$$\bigcap_{x \in [0,1]} [x,1] \times [0,x^2] = \{(1,0)\}.$$

Chapter 3.5, Question 16:

Suppose $A = \{1,2,\ldots,10\}$. Notice that A has 5 even elements and 5 odd elements. To have exactly 3 even elements in a 6-element subset means choosing 3 even and 3 odd. That is, the number of 6-element subsets of A with exactly 3 even elements is

$$\binom{5}{3}\binom{5}{3} = 100.$$

There are $\binom{10}{6} = 210$ 6-element subsets of A. So the number of them that don't have exactly 3 even elements is $210 - 100 = 110$.

Chapter 3.10, Question 6:

Let n be a non-negative integer. Consider $\binom{3n}{3}$. This is the number of 3-element subsets of a set S of size $3n$. Partition S into $A \cup B \cup C$, where $|A| = |B| = |C| = n$.

Every 3-element subset of S falls into exactly 1 of these 3 cases:

Case A: All 3 elements are chosen from the same subset, either from A, B, or C. So there are 3 options of subsets and $\binom{n}{3}$ ways to choose elements. Thus, we have $3\binom{n}{3}$ distinct 3-element subsets.

Case B: We choose 2 elements from one subset and 1 element from another subset. Then, there are $\binom{3}{1}$ ways to choose the subset with 2 elements, $\binom{2}{1}$ ways to choose the other subset with 1 element, and $\binom{n}{1}\binom{n}{2}$ ways to choose elements. Thus, by multiplication principle, we have an additional $6n\binom{n}{2}$ distinct 3-element subsets.

Case C: We choose 1 element from each of the subsets. That is, we have $\binom{n}{1}^3$ ways to choose the elements. Thus, by multiplication principle, we have another n^3 distinct 3-element subsets.

Since all the cases are disjoint and completely cover all 3-element subsets, it follows that we have a total of $3\binom{n}{3} + 6n\binom{n}{2} + n^3$ distinct 3-element subsets from S.

Both of these expressions count the same subsets of S. So we conclude

$$\binom{3n}{3} = 3\binom{n}{3} + 6n\binom{n}{2} + n^3.$$

Chapter 8, Question 2:

Let A denote $\{6n : n \in \mathbb{Z}\}$, B denote $\{2n : n \in \mathbb{Z}\}$, and C denote $\{3n : n \in \mathbb{Z}\}$.

Suppose $a \in A$. That is, $a = 6k$ for some integer k. Then,

$$\begin{aligned} a &= 2(3k) \\ &= 2w, \text{ where } w = 3k. \end{aligned}$$

Then, w is an integer since it's the product of integers. So $a \in B$. We also have

$$\begin{aligned} a &= 3(2k) \\ &= 3z, \text{ where } z = 2k. \end{aligned}$$

Then, z is an integer since it's the product of integers. So $a \in C$. Since a is in both B and C, it follows that $a \in B \cap C$. Thus, $A \subseteq B \cap C$.

Suppose $d \in B \cap C$. That is, for some integers x and y, we have $d = 2x$ and $d = 3y$. Then, $2x = 3y$. So $3 \mid 2x$. Since 3 is prime and $gcd(2, 3) = 1$, by Euclid's Lemma, we have $3 \mid x$. That is, for some integer n, $x = 3n$. Then,

$$\begin{aligned} d &= 2(3n) \\ &= 6n. \end{aligned}$$

So $d \in A$. Thus, $B \cap C \subseteq A$.

Since $A \subseteq B \cap C$ and $B \cap C \subseteq A$, we conclude $A = B \cap C$. That is,

$$\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}.$$

Chapter 8, Question 16:

Let A, B, and C be sets.

Let $x \in A$ and $y \in B \cup C$. Then, $(x, y) \in A \times (B \cup C)$. Since y is either in B or C, we have 2 cases for y.

Case 1: $y \in B$. So $(x, y) \in A \times B$.

Case 2: $y \in C$. So $(x, y) \in A \times C$.

Thus, $(x, y) \in (A \times B) \cup (A \times C)$. Then, $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$.

Let $(a, d) \in (A \times B) \cup (A \times C)$. Then, we have the following 2 cases:

Case 1: $(a, d) \in (A \times B)$. So $a \in A$ and $d \in B$.

Case 2: $(a, d) \in (A \times C)$. So $a \in A$ and $d \in C$.

Notice that a is always in A, while d is either in B or C. That is, $a \in A$ and $d \in B \cup C$. So $(a, d) \in A \times (B \cup C)$. Thus, $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$.

Therefore, $(A \times B) \cup (A \times C) = A \times (B \cup C)$.

Chapter 8, Question 28:

Let X denote $\{12a + 25b : a, b \in \mathbb{Z}\}$.

Suppose $x \in X$. Then, $x = 12a + 25b$ for some integers a and b. Since x is a sum of products of integers, x is an integer. So $x \in \mathbb{Z}$. Thus, $X \subseteq \mathbb{Z}$.

Suppose $z \in \mathbb{Z}$. Consider $12a + 25b$. Let $a = -2z$ and $b = z$. Since z is an integer, a and b are also integers. Then,

$$12(-2z) + 25(z) = z.$$

Thus, we have found a and b such that $12a + 25b = z$. So, $z \in X$. Thus, $\mathbb{Z} \subseteq X$.

Since we have $X \subseteq \mathbb{Z}$ and $\mathbb{Z} \subseteq X$, we can conclude that $X = \mathbb{Z}$. That is,

$$\{12a + 25b : a, b \in \mathbb{Z}\} = \mathbb{Z}.$$